

# PLANNING, SECURITY, AND YOUR FIRST REPOSITORY ENTRY

Submission Form

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|-------------------------------------------------------|---------------------------------------------------|
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| <b>Course/Section:</b> <u>BSIT-IV 80107</u>           | <b>Date:</b> <u>September 2, 2026</u>             |

**Chosen App or Organization (semester project):** AI-Powered Campus Helpdesk and Ticket Routing System  
Using Multi-Label Text Classification

## Part 1. Planning Question Evaluation

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**New request being evaluated (one sentence):**

The organization wants to add an email notification feature that automatically notifies students when their helpdesk ticket is received, routed, updated, or resolved.

### 1. Does this solve a real problem?

Yes, students may not always know whether their submitted ticket has been received or what department is currently handling it. Email notifications would provide timely updates and reduce the need for students to repeatedly check the helpdesk system.

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### 2. Do we already have something that can do this?

The existing helpdesk system already stores ticket information and tracks ticket status, but it does not automatically send notifications to students. Therefore, an email notification component would extend the existing system rather than duplicate an existing feature.

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### 3. Does this support the organization's actual goals?

Yes, the feature supports the goal of improving communication between students and faculty or staff. It can also improve transparency and user satisfaction by keeping students informed about the progress of their concerns.

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### 4. Is it secure and cost-effective?

The feature can be implemented securely by using authenticated email services and by sending only necessary ticket information. It is also relatively cost-effective because it can be integrated with the existing helpdesk system without requiring a completely separate application.

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**Where would this rank against other approved work? (value, risk, dependency, or effort, reasoned in words, not calculated)**

This request would likely rank high to medium in value because it directly improves communication and user experience while requiring relatively low development effort. Its security risk and dependency are manageable because it can use the existing authentication, ticket, and database infrastructure.

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## Part 2. Security Pass Across the Four Domains

| Domain      | Security Concern                                                                                                                                    | Matching Control                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business    | Unauthorized users may access helpdesk functions that are intended only for students, faculty/staff, or administrators.                             | Implement role-based access control (RBAC) so users can access only the functions appropriate to their assigned roles.                                               |
| Data        | Student information and ticket details may be exposed to users who are not authorized to view them.                                                 | Use database access policies and encryption to restrict access to sensitive records and protect data during storage during transmission.                             |
| Application | Attackers or unauthorized users could manipulate ticket information or submit malicious input through the helpdesk forms.                           | Apply input validation, authentication, and secure API/database queries throughout the application.                                                                  |
| Technology  | The application's backend services and database could be compromised because of outdated software, exposed credentials, or insecure configurations. | Keep dependencies and services updated, protect API keys and credentials using environment variables/secrets management, and enforce secure configuration practices. |

**One weak-link risk (how a weak point in one part could put a stronger part at risk):**

A weakly protected user account could allow an attacker to access the helpdesk system and potentially reach sensitive ticket or student data even if the database and other components have strong security controls.

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## Part 3. Your First Repository Entry

**Component being documented (application, data store, or technology platform):** Supabase

| Field                                  | Your Entry                                                                                                                                                                                                                |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What it does                           | The Supabase database stores the system's user accounts, roles, helpdesk tickets, ticket categories, routing information, feedback, and ticket status records.                                                            |
| Which department or user group uses it | Students, faculty/staff, and administrators use the data stored in the database through the helpdesk application. Administrators and authorized personnel may have additional access for managing and monitoring tickets. |
| What it connects to                    | It connects to the React/Vite frontend, authentication system, AI-based ticket classification and routing process, and other application services through the application's backend/API layer.                            |
| Who is responsible for it              | The system development team is responsible for database design, configuration, maintenance, security policies, and data integrity, with authorized system administrators overseeing its operational use.                  |

| Field                               | Your Entry                                                                                                                                                                                                   |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Whether it is still needed, and why | Yes, it is still needed because the helpdesk system depends on persistent storage for users, tickets, routing results, feedback, and other records required for the system to function and generate reports. |